Ans. (a) : Ultraviolet radiation in water purification neutralizes/destroys harmful micro-organism present in water. Certain gases located in the atmosphere (i.e. Ozone) are allowed to come to the ground only after filtering the earth radiation waves. The gases are present the ultraviolet radiation of the Sun in the middle and protect the surface from overheating.

75. A team of scientists at Brookhaven National laboratory including those from India created the heaviest anti-matter (anti-helium nucleus).

What is/are the implication/implications of the creation of anti-matter ?

1. It will make mineral prospecting and oil exploration easier and cheaper.
2. It will help probe the possibility of the existence of stars and galaxies made of anti matter.
3. It will help understand the evolution of the universe.

Select the correct answer using, the codes given below :

(a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans. (b): The scientific team of the Brookhaven National Laboratory, which also included Indian scientists, produced the heaviest anti-matter in the form of anti-helium. This will be helpful in checking the possibility of stars or galaxies formed from anti-matter and it will also be helpful in developing an understanding of the universe.

• Botany

76. Consider the following kinds of organisms:

1. Bat
2. Bee
3. Bird

Which of the above is/are pollinating agent/ agents?

(a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d): Pollination, in the scientific sense, is the process of pollen reaching the ovary or spore, which produces fruits and seed after conception. Bees acts as pollinators as well as birds in tropical regions are also commonly known as pollinators. Bats are also pollinators of many flowers.

• Medical Science, Anthropology

77. Consider the following kinds of organisms :

1. Bacteria
2. Fungi
3. Flowering plants

Some species of which of the above kind of organisms are employed as biopesticides ?

(a) 1 only (b) 2 and 3 only
(c) 1 and 2 only (d) 1, 2 and 3

Ans. (d): A fungal biopesticide formulation has been developed by a faculty member of the department of Biotechnology at Gitam University and the test results held at different districts showed that it could control several pests affecting chilli, paddy and vegetable crops.

78. Lead, ingested or inhaled is a health hazard. After the addition of lead to petrol has been banned, what still are the sources of lead poisoning?

1. Smelting units
2. Pens and pencils
3. Paints
4. Hair oils and cosmetics

Select the correct answer using the codes given below :

(a) 1, 2 and 3 only (b) 1 and 3 only
(c) 2 and 4 only (d) 1, 2, 3 and 4

Ans. (b): Among the options in questions are sources of lead poisoning are smelting units and paints. The amount of lead found in paint used for painting in homes is fatal to human health. The welding/smelting units of zinc and copper release lead into the atmosphere.

79. With reference to 'stem cells', frequently in the news, which of the following statements is / are correct?

1. Stem cells can be derived from mammals only.
2. Stem cells can be used for screening new drugs.
3. Stem cells can be used for medical therapies.

Select the correct answer using the codes given below :

(a) 1 and 2 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans. (b): 'Stem cells', often in the spotlight can be used to test new drugs. In addition, they can also be used for medical therapy. It can independently perform such actions as nutrition, respiratory enhancement, reproduction and self-regulation.

• Agriculture, Animal Husbandry and Dairy

80. Consider the following :

1. Assessment of land revenue on the basis of nature of the soil and the quality of crops.
2. Use of mobile cannons in warfare.
3. Cultivation of tobacco and red chillies.

Which of the above was/were introduced in to India by the English?

(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) None

Ans. (d): Land revenue was assessed only in the medieval period based on the nature of the soil and the properties of the produce. In the war, mobile canons were used by Babur in the first battle of Panipat. Apart from this, tobacco was first brought to India by the Portuguese during the reign of Mughal Emperor Akbar.

81. How does National Biodiversity Authority (NBA) help in protecting the Indian agriculture ?

1. NBA checks the biopiracy and protects the indigenous and traditional genetic resources.
2. NBA directly monitors and supervises the scientific research on genetic modification of crop plants.
3. Application for Intellectual Property Rights related to genetic/biological resources cannot be made without the approval of NBA .

Which of the statements given above is /are correct ?

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c): An application for intellectual property right related to genetic/biological resources cannot be applied without the recommendation of the National Biodiversity Authority. This authority prohibits bio-theft and treats native and traditional genetic resources. The Genetic Engineering Appraisal Committee (GEAC) is responsible for approval of proposal relating of Genetically Engineered organisms and products into the environment including experimental field trials as per the provisions of rules 1998.

• Science and Technology

82. What are the reasons for the people's resistance to the introduction of Bt brinjal in India?

1. Bt brinjal has been created by inserting a gene from a soil fungus into its genome.
2. The seeds of Bt brinjal are terminator seeds and therefore, the farmers have to buy the seeds before every season from the seed companies.
3. There is an apprehension that the consumption of Bt brinjal may have adverse impact on health.
4. There are some concerns that the introduction of BT brinjal may have adverse effect on the biodiversity.

Select the correct answer using the codes given below:

(a) 1, 2 and 3 only (b) 2 and 3 only
(c) 3 and 4 only (d) 1, 2, 3 and 4

Ans. (c): The main reason for people opposing the introduction of Bt-Brinjal in India is that introducing Bt brinjal will have an adverse effect on biodiversity and possibility that the use of Bt brinjal may have adverse effects on health. Apart from this, the farmers will also have the opposite effect on the labourers working in fields and other tree plants.

83. Other than resistance to pests, what are the prospects for which genetically engineered plants have been created ?

1. To enable them to withstand drought.
2. To increase the nutritive value of the produce.
3. To enable them to grow and do photosynthesis in spaceships and space stations.
4. To increase their shelf life.

Select the correct answer using the codes given below:

(a) 1 and 2 only (b) 3 and 4 only
(c) 1, 2 and 4 (d) 1, 2, 3 and 4

Ans. (c) : All statements are correct. Theoretically, there are no limits to the extent of modification produced by genetic engineering.

84. Biomass gasification is considered to be one of the sustainable solutions to the power crisis in India. In this context, which of the following statements is/are correct ?

1. Coconut shells, groundnut shells and rice husk can be used in biomass gasification .
2. The combustible gases generated from biomass gasification consist of hydrogen and carbon dioxide only.
3. The combustible gases generated from biomass gasification can be used for direct heat generation but not in internal combustion engines.

Select the correct answer using the codes given below:

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (a): Biomass gasification is considered one of the sustainable solutions to the energy crisis. In this context, coconut shell, peanut peel and paddy husk are used for biomass gasification.

85. Consider the following statements :

Chlorofluorocarbons, known as ozone-depleting substances, are used-

1. in the production of plastic foams.
2. in the production of tubeless tyres.
3. in cleaning certain electronic components.
4. as pressurizing, agents in aerosol cans.

Which of the statements given above is /are correct?

(a) 1, 2 and 3 only (b) 4 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

Ans. (c): Chloroflurocarbons, popularly known as ozone depleting substances are used to clean certain electronic components. Aerosol is used as a compressive agent in cars and also in the manufacture of solvable foam. CFCs are simple compounds of chlorine, fluorine and carbon.

ENVIRONMENT AND ECOLOGY

86. The increasing amount of carbon dioxide in the air is slowly raising the temperature of the atmosphere, because it absorbs:

(a) the water vapour of the air and retains its heat.
(b) the ultraviolet part of the solar radiation.
(c) all the solar radiations.
(d) the infrared part of the solar radiation.

Ans. (d): Among the Greenhouse gases (GHGs), only water vapour has the ability to absorb both incoming (UV) and outgoing (infrared) radiation.

87. In which one among the following categories of protected areas in India are local people not allowed to collect and use the biomass ?

(a) Biosphere Reserve
(b) National Park
(c) Wetlands declared under Ramsar Conventions
(d) Wildlife Sanctuaries